

Chapter 3

Methodology and Requirements Analysis

3 Methodology and Requirements Analysis

This chapter summarizes the *methodology that will be used during the development of this project* and *offers an analysis of the project requirements necessary* to ensure its successful implementation. The methodology provides a structured framework to guide each phase of development from initial planning and data gathering to design, implementation, and evaluation. In addition, this chapter presents a detailed requirements analysis, which is important in identifying the key functionalities, performance expectations, and constraints of the system. This includes both functional requirements and non-functional requirements, which address how the system should perform. The requirements were gathered using various techniques to ensure they accurately reflect the needs of the users. Together, the methodology and requirements analysis serve as a foundation for the system's design and development.

3.1 Methodology

Methodology is a structured approach or framework that guides how projects are planned, executed, monitored, controlled, and closed (GeeksforGeeks, 2024). The reason that methodology is important is that it provides a *standardized approach* to managing undertakings, *ensuring consistency and repeatability across different projects* within an organization (GeeksforGeeks, 2024). Additionally, it *helps align stakeholders and project teams* by clearly defining project objectives, roles, responsibilities, and expectations (GeeksforGeeks, 2024). Furthermore, methodology is also *flexible* in adapting to changes in project requirements, environments, and constraints (GeeksforGeeks, 2024). There are multiple selections of methodologies that can be implemented in this project, such as Waterfall, SCRUM, and Agile. It is crucial to choose the right methodologies because they can affect the success of a project (GeeksforGeeks, 2024). Therefore, after evaluating the project's scope and requirements, *Extreme Programming (XP)* has been selected as the most suitable approach for the Artist E-commerce system.

3.1.1 Proposed Methodology

XP is an *Agile software development* methodology that focuses on delivering high-quality software through frequent and continuous feedback, collaboration, and adaptation (GeeksforGeeks, 2025a). *XP emphasizes a close working relationship* between the development team, the customer, and stakeholders, with an emphasis on rapid, iterative

development and deployment (GeeksforGeeks, 2025a). XP operates through a set of iterative and collaborative stages designed to ensure high-quality software development that remains closely aligned with customer needs.

3.1.2 Software Development Life Cycle of Extreme Programming

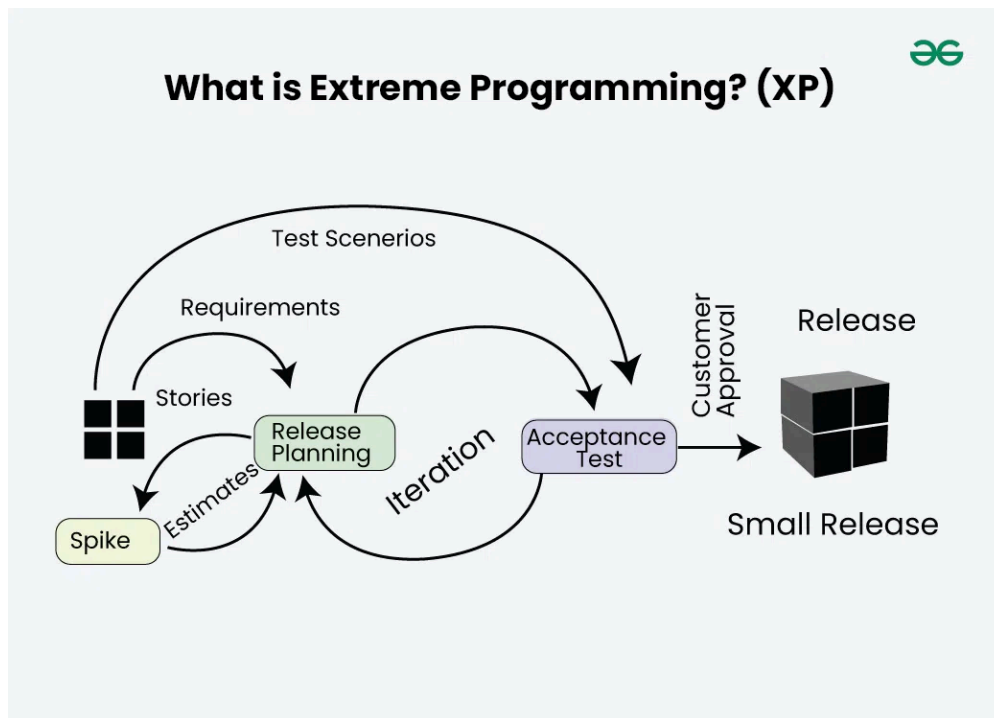


Figure 3.1: Software Development Life Cycle of Extreme Programming (XP) Diagram

Source:

<https://www.geeksforgeeks.org/software-engineering/software-engineering-extreme-programming-xp/>

The **XP Life Cycle** consists of five stages. The first stage of XP is the planning stage, where customers define their needs in concise descriptions known as **user stories** to form the basis of the project's goal (GeeksforGeeks, 2025a). These stories will then be **calculated** to **determine the effort required for each story** and **schedule releases** according to priority and effort (GeeksforGeeks, 2025a). After that, these stories will be utilized on the **Design stage** to develop the code architecture to **meet the customer's expectations** (Suresh Konduru, 2024). This includes deciding on the **programming language, environment, libraries, and frameworks** to be used in software development to help keep the design straightforward and clear (Suresh Konduru, 2024). Then, in the **Coding stage**, the development team writes the actual code. The code will be **written in small chunks and tested regularly** to ensure that the

code is high quality and that any bugs can be addressed quickly (Suresh Konduru, 2024). They *write tests before coding* to ensure functionality from the start, and frequently integrate their code into a shared repository with automated tests to catch issues early (GeeksforGeeks, 2025a). After that, in the *Testing stage*, XP places strong focus on continuous testing through both *unit tests* and *acceptance tests* to ensure features work correctly and ensure that the overall system meets initial requirements (GeeksforGeeks, 2025a). Lastly, in the *Listening stage*, the customer *gives feedback* on the software and provides input that potentially can make the system better. This helps ensure that the software meets the customer's expectations and is of the highest quality (Suresh Konduru, 2024).

3.1.3 Justification for Using Extreme Programming (XP) Methodology

The reason that XP is a *suitable methodology* for the Artist E-commerce system is that it can *adapt to changing requirements*. XP is designed to *handle rapidly changing requirements*, and thus it suits Artist E-commerce since the platform has a tendency to evolve frequently based on users' reviews, new trends in art, and market needs (GeeksforGeeks, 2025a). Another reason that XP is a suitable methodology is that it *encourages collaboration between developers, customers, and stakeholders* (GeeksforGeeks, 2025a). Customers are *included as part of the team* and involved in every iteration, providing feedback and detailing requirements. This ensures that the final project *reflects the actual user needs* rather than assumptions made at the beginning of the project. Moreover, XP *highly values the quality of software*, where it places a very *high emphasis on quality assurance and testing* (GeeksforGeeks, 2025a). It integrates practices such as *Test-Driven Development (TDD)* and *frequent refactoring* to help *detect and fix defects early*, *reduce technical debt*, and ensure that the system remains *maintainable* as it evolves (GeeksforGeeks, 2025a). For an Artist E-commerce system, this high standard of quality is important because it provides a bug-free experience for both artists and customers.

3.2 Requirements Gathering Technique

3.2.1 Document Analysis

Document analysis is a qualitative research technique that evaluates electronic and physical documents to interpret them, gain an understanding of their meaning, and develop upon the information they provide (Indeed Editorial Team, 2025). To identify *key user requirements*

for the Artist E-Commerce System, **9 journals and 1 thesis published between 2022 and 2025 were gathered and reviewed**. These sources cover technologies and user-centered needs across five key modules: Sales, Purchase, AI Chatbot, Artwork Fraud Detection, and Payment. A total of 10 documents were analyzed.

1. This journal explores how the **language style** of AI-powered customer service chatbots (formal vs. informal) affects users' **continued usage intention** and **attitude toward the brand** in an e-commerce context (Meichan Lee & Rui Wang, 2023). The results show **that an informal language style** significantly enhances the **parasocial interaction**, a sense of human-like connection between users and chatbots (Meichan Lee & Rui Wang, 2023). It mentions that **AI Chatbot** has **a 45% increase in usage** during 2020, and it is expected to **undertake more than 60% of customer service** by 2023 (Meichan Lee & Rui Wang, 2023). Despite their growing presence, some customers still prefer **human interaction**.
2. A journal by **Majid Mirbod (2024)** introduces an **applied machine-vision-based model** for artwork authenticity recognition, leveraging **image processing, machine vision**, and a **fuzzy logic interface**. The journal emphasizes that verifying an artwork's authenticity is really challenging, as artworks are frequently targeted by counterfeiters due to their **artistic value, uniqueness**, and **financial worth** (Majid Mirbod, 2024). Therefore, the author decided to propose a system that **compares high-resolution images of original artworks** with those of **potentially forged copies** (Majid Mirbod, 2024). This comparison allows the model to detect even the slightest visual differences. The system relies on having a verified original image on record, making it a **reference-based verification approach**. Tested on seven globally renowned artworks, the model proved both **accurate and reliable** (Majid Mirbod, 2024).
3. A journal by **Koerniawan Hidajat and S. Sarwita Salini (2023)** highlights that payment gateways are crucial for **secure and efficient e-commerce transactions**. They provide **encrypted transaction processing**, helping protect sensitive financial data from fraud. The system supports **multiple payment methods** like credit cards, digital wallets, and bank transfers, which increases flexibility and convenience for users (Koerniawan Hidajat and S. Sarwita Salini, 2023). Additionally, payment gateways enable **real-time processing, which reduces checkout time** and enhances

the *user experience* (Koerniawan Hidajat and S. Sarwita Salini, 2023). Features such as *transaction tracking*, *analytics*, and *automated receipts* simplify both seller and customer operations.

4. This journal by *Jingyun Zhange (2023)* proposes a *recommendation system* for online art platforms that enhances the *personalization of artwork suggestions* (Jingyun Zhange, 2023). It uses a **user keyword-driven correlation graph search** technique, where the system constructs a graph linking users, keywords, and artworks based on browsing history, preferences, and contextual relevance (Jingyun Zhange, 2023). The recommendation is *composite*, meaning it combines various factors, such as user intent, semantic similarity of keywords, and artwork attributes like the style, theme, and medium to deliver *more accurate and tailored results* (Jingyun Zhange, 2023). The system improves both *user satisfaction* and *purchase likelihood*, especially in digital art marketplaces where users often struggle to find pieces that match their taste (Jingyun Zhange, 2023).
5. This journal about "E-commerce Websites for Artisans" by *Prof. S.C. Shirbhate (2023)* explores the development of an *e-commerce platform tailored for artists*, focusing on how such a system can empower creators to *sell handmade, customized, or artistic products* directly to consumers (Prof. S.C. Shirbhate, 2023). The study highlights that many artisans face challenges in traditional marketplaces, such as *limited reach*, *lack of digital tools*, and *difficulties managing payments and inventory* (Prof. S.C. Shirbhate, 2023). The proposed system includes modules for *product listing*, *cart management*, *secure payment integration*, *product search and filtering*, *reviews and rating*, and *order tracking*, with special attention to *usability and visual presentation* of artisanal goods (Prof. S.C. Shirbhate, 2023). The platform encourages *artists to showcase* their product uniqueness through detailed descriptions and visuals to boost customer connection and trust (Prof. S.C. Shirbhate, 2023).
6. The journal by *M. Shyam Manikanta (2022)* describes the architecture and development of a web-based shopping platform improved with chatbot capabilities. The system integrates core e-commerce features, including *user registration*, *product browsing*, *shopping cart functionality*, and *payment workflows* with an intelligent chatbot named JAICOB (M. Shyam Manikanta, 2022). The chatbot *automates*

customer service by *interpreting natural language queries*, *matching user input against a pre-defined list* using Levenshtein distance and AIML-based rules, and facilitating actions such as *inventory checks*, *order placement*, and *basic payment guidance* (M. Shyam Manikanta, 2022). This integrated design improves user engagement by offering *real-time conversational support* throughout the shopping experience, *reducing manual workload for sellers* while providing immediate query resolution (M. Shyam Manikanta, 2022). The chatbot serves as a virtual assistant by answering commonly asked questions and guiding users through product selection and checkout.

7. The journal by *Ensuk Yeong, Wong Young Chong, and Daeho Lee (2023)* studies how *consumer trust in product quality* is influenced by *online reviews and the design of e-commerce platforms*. The study emphasizes that trust is significantly impacted by the *credibility, quantity, and tone of reviews*, especially when they include *images or detailed user experiences*. Reviews from buyers *carry more weight and positively influence purchase decisions*. Additionally, the *platform's reputation, layout clarity, secure payment features, and transparent return policies* contribute to building consumer confidence. The findings suggest that integrating a *review system* and maintaining a *trustworthy shopping interface* are crucial for online sellers, especially in markets involving artistic or high-value products.
8. The journal by *Suraj Basnet (2024)* investigates the *adoption of e-commerce by Finnish handicraft artisans and consumers*, highlighting both *advantages and challenges* that may occur when using an e-commerce platform (Suraj Basnet, 2024). Consumers value the *convenience, variety, and accessibility* of online shopping, while artisans benefit from *wider market exposure* and *direct-to-consumer sales* (Suraj Basnet, 2024). However, many artisans face difficulties such as *limited digital skills, technology barriers*, and challenges in *conveying product authenticity online* (Suraj Basnet, 2024). This study highlights the role of *storytelling, high-quality visuals, and customer feedback* in establishing consumer trust and enhancing the online shopping experience for handmade products.
9. The journal by *Andrii Martynenko (2022)* explores the challenges involved in verifying the authenticity of artworks. It highlights that determining whether a

painting is genuine involves analyzing multiple factors such as *artistic technique, materials used, historical data, and provenance documentation* (Andrii Martynenko, 2022). This study also mentions that the *lack of standardized systems* in the art industry often leads to *uncertainty about an artwork's originality* (Andrii Martynenko, 2022). Therefore, this journal suggests implementing a *systematic, technology-aided approach*, including image analysis and databases, to *enhance the credibility and efficiency* of authenticity verification (Andrii Martynenko, 2022). These findings support the integration of a *fraud detection or artwork verification module* to ensure that listed artworks are original and protect artists and buyers from counterfeit products.

10. The thesis of *Top E-Commerce CMS Platforms* by *Seunghun Bang (2023)* provides a comprehensive comparison of the most widely used content management systems for building online stores, focusing on *Shopify, WordPress with WooCommerce, and BigCommerce*. These platforms offer essential tools and features required to establish a successful e-commerce presence, such as *product management, payment gateway integration, inventory tracking, responsive design, and built-in SEO tools* to enhance visibility and user engagement (Seunghun Bang, 2023). *Shopify* is recognized for its *ease of use, hosted environment, and simplified setup process*, making it ideal for individuals or small businesses with limited technical expertise (Seunghun Bang, 2023). It comes with built-in features like *payment processing, themes, and analytics*, allowing users to launch their online store quickly without dealing with backend infrastructure (Seunghun Bang, 2023). On the other hand, *WordPress with WooCommerce* provides a highly customizable and *flexible solution*. It benefits from WordPress's vast plugin ecosystem, giving users *greater control over the design and functionality* of their store (Seunghun Bang, 2023). This is useful for artists who want to tailor their website to reflect their unique brand identity. *BigCommerce combines user-friendliness with powerful enterprise-level features*, making it suitable for growing businesses that *need scalability and integration with other sales channels* like Amazon, eBay, and social media (Seunghun Bang, 2023). It offers strong support for multi-channel retailing, detailed analytics, and security features.

3.2.2 System Review

System review is a technique used to *evaluate and assess the existing system modules and their functions*. It helps determine how well each component performs in meeting the intended goals and user requirements. For this project, **5 existing systems** related to Artist E-commerce platforms were reviewed to gain insights into the core modules typically used in the industry. These modules include the *Sales Module, Purchase Module, Payment Module, AI Chatbot Module, and Artwork Fraud Detection Module*.

1. **Etsy** is a widely recognized online marketplace tailored for handmade and artistic goods. Its **sales module** allows artists to easily *upload artwork, set prices, manage inventory, and categorize listings*. The **purchase experience** is user-friendly, enabling consumers to *browse products, add items to a cart, and checkout* without confusion. Etsy supports a variety of *payment methods*, including credit cards, PayPal, Apple Pay, and more, making it flexible for international buyers. It also provides a **built-in AI chatbot** for *customer service*, helping users with common questions, such as *order tracking*. Etsy operates its own **fraud detection system** to identify *suspicious transactions or account behavior*, helping maintain a secure environment for both consumers and artists. Overall, Etsy is ideal for individual artists seeking global exposure with a relatively simple setup, though *transaction fees and competitive visibility* may pose challenges.
2. **Shopify** is a powerful and flexible e-commerce platform often used by artists who want to build their e-commerce platform. Its **sales module** is customizable, where artists can add features like *uploading artwork, managing real-time inventory, setting prices, and organizing promotions* with ease. The **purchase experience** can be tailored to match the artist's branding, providing a fast and secure checkout process. Shopify supports various *payment methods*, including Shopify Payments, PayPal, Stripe, and Apple Pay, making it accessible for a global audience. For customer interaction, Shopify supports *third-party AI chatbot integrations*, such as Tidio, Gorgias, or ChatBot.com, offering *automated responses* in customer service, FAQs, and even abandoned cart reminders. Shopify does not have any built-in *artwork fraud detection system*, but artist can still install a *third-party artwork fraud detector* on their platform where they are responsible for protecting their artwork. Its biggest advantage is control, ideal for artists who want *ownership over their platform*. However, it does require some *technical effort* and *subscription fees* to use Shopify

features.

3. **ArtStation** functions as both a *professional portfolio platform* and an *online marketplace*, making it especially well-suited for digital and traditional artists. Its **sales module** allows creators to list and sell digital downloads, physical prints, and original artworks through a *user-friendly interface*. The **purchase module** is simplified for *easy browsing and discovery*, catering to both art enthusiasts and industry professionals. For **payment**, ArtStation supports *secure and reliable* gateways such as PayPal and Stripe, ensuring smooth transactions. While it doesn't include a built-in *AI chatbot*, artists can integrate third-party services if needed, though most communication is handled through direct messaging. When it comes to *artwork fraud detection systems*, ArtStation relies on *manual review processes* and *content moderation* to *maintain quality and prevent misuse*. This approach is effective for ensuring professional standards but may not scale well for high-volume or automated enforcement. Overall, ArtStation excels in serving concept artists, illustrators, and digital creators, though it offers fewer tools for selling physical goods compared to broader e-commerce platforms.
4. **Redbubble** is a popular *print-on-demand marketplace* that enables artists to sell their artwork on a wide variety of products such as *clothing, stickers, home décor, and accessories*. The **Sales Module** allows artists to easily *upload their designs, set pricing margins*, and choose which *products to sell*. It *handles the production, packaging, and global shipping*, allowing artists to focus on creativity in their art. On the buyer side, the **Purchase Module** is *user-friendly*, offering *intuitive browsing* by category, style, and trending content, with quick access to reviews and artist profiles. Redbubble's **Payment Module** supports *multiple payment options*, including credit/debit cards, PayPal, and other region-specific processors, and artists are paid monthly based on their earnings. While Redbubble does not offer a built-in **AI chatbot**, it *provides automated help center responses* and *email-based support*, which cover common queries but may lack real-time interaction. In terms of **Artwork Fraud Detection**, Redbubble uses a *combination of manual moderation and community reporting* to address art theft and copyright violations. However, the platform has faced ongoing criticism from artists about the ease with which stolen designs can be uploaded by others, highlighting limitations in its *proactive protection*.

methods. Despite this, Redbubble remains a powerful tool for artists seeking passive income and global exposure, especially for digital creators who prefer hands-off fulfillment.

5. **DeviantArt** is one of the largest online communities for artists, offering both *portfolio features* and *marketplace functionalities*. Its **Sales Module** allows artists to *sell digital downloads and prints* of their artwork through DeviantArt's built-in "Prints" program, where artists can *set prices* and *choose product formats*. While not as comprehensive as full e-commerce platforms, it provides a simple way to *monetize visual creations*. The **Purchase Module** is integrated directly into the artwork display pages, enabling consumers to quickly *view available print options*, *select sizes*, and *make purchases*. The **Payment Module** uses a *combination of DeviantArt Points (a virtual currency)* and *payment gateways* like PayPal for artists to receive earnings, though the Points system can be confusing for newcomers. DeviantArt does not include a dedicated *AI chatbot*, but it does *feature a help center and support system* that relies on tickets and documentation rather than live assistance. On the **Artwork Fraud Detection** side, DeviantArt employs *content moderation and community-driven reporting* to flag stolen or misused artworks. Artists can submit DMCA takedown requests, and DeviantArt also allows *watermarking of uploads to deter theft*. Despite some limitations in automation and payment clarity, DeviantArt remains a strong platform for both showcasing and selling digital art, particularly for creators interested in building community alongside commerce.

3.2.3 Result

Based on the above fact collected for document analysis and system reviews, there are some requirements and functions that are *commonly aligned* across both academic research and real-world implementations of artist e-commerce platforms. These findings serve as a *valuable reference* for the development of our Artist E-Commerce system, ensuring that it incorporates all major features observed in existing platforms and validated by recent scholarly research.

Customizable Sales Module

Inspired by Etsy, Shopify, and findings from Prof. Shirbhate (2023), our system enables artists to *upload artworks* with detailed descriptions, *manage inventory*, *set pricing*, and *personalize artwork listings* using categories, tags, and high-quality visuals.

Personalized Artwork Recommendations

Based on the model proposed by Jingyun Zhange (2023), the platform will implement a *keyword-driven recommendation system* that links user preferences with artwork style, medium, and theme to increase engagement and sales conversion.

Smooth Purchase Experience

Reflecting best practices from Redbubble, DeviantArt, and user trust studies by Ensuk Yeong (2023), our purchase module includes *easy navigation*, *product filtering*, *secure checkout*, *customer reviews*, and *wishlist functionality* to support buyer confidence.

Intelligent AI Chatbot Support

Following trends in Lee & Wang (2023) and the chatbot model in M. Shyam Manikanta (2022), our system will integrate a *natural-language chatbot* that provides customer service to consumers, *FAQ support*, and *escalation to human agents when needed*.

Artwork Fraud Detection

Informed by Mirbod (2024) and Martynenko (2022), our system incorporates an *image analysis system* that *compares uploaded artworks with verified originals*, alongside an option for *watermarking artworks and community reporting features* to combat counterfeit artworks.

Secure and Flexible Payment Gateway

Studies from Hidajat & Salini (2023) and practices in Shopify and ArtStation, the payment module will *support multiple payment options*, including credit/debit cards, e-wallets, and bank transfers, along with real-time processing and analytics for both artists and customers.

3.3 Requirement Analysis

Requirement Analysis is a process where the needs and expectations of users are *identified and documented* (Shweta Jain, 2025). It serves as a foundation for the development process,

where it ensures that the final system *meets user expectations* and business goals (Shweta Jain, 2025). These requirements will be *analyzed and transformed into visual representations* such as *use case diagrams* for the Artist E-commerce systems. These diagrams help developers visualize system functionality, making complex requirements easier to understand and implement.

3.3.1 Overview Use Case Diagram

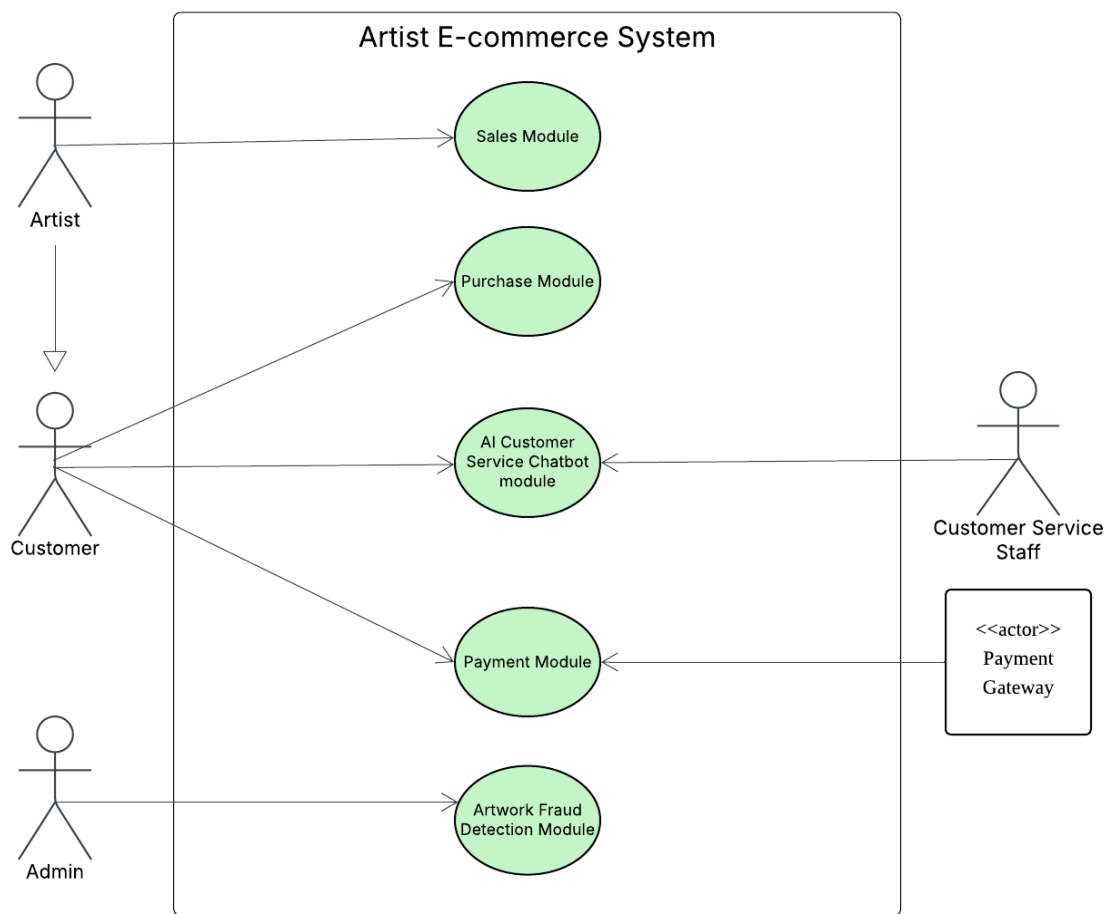


Figure 3.2 Artist E-commerce Overview Use Case Diagram

Based on Figure 3.0, the *overall Use Case Diagram of the Artist E-commerce* system shows how various actors interact with key modules in the system. There are a total of *five actors* in this system, which are the *artist*, *customer*, *admin*, *customer service staff*, and *payment gateway*. There are also a total of *five modules* that are included inside the diagram, which are the *Sales Module*, *Purchase Module*, *Artwork Fraud Detection Module*, *Payment Module*, and *AI Customer Service Chatbot Module*.

The **Sales module** is *interacted with by the artist* to upload and manage their artwork listing. The **Purchase module** is *used by the customer* to browse and purchase artwork available in the artwork product list. Additionally, **customer** interacts with the **AI Customer Service Chatbot module** to inquire about any concerns, like product delivery status. The **customer service staff** also engages with the **AI Customer Service Chatbot module** to handle more complex questions and provide live support when the customer requests it. **The Payment module** is *used by the customer* after the checkout process to securely complete their transactions, which are processed through an external **Payment Gateway**. Meanwhile, the **Artwork Fraud Detection Module** is managed by the **Admin** to monitor and prevent fraudulent artwork listings, ensuring the authenticity and trustworthiness of the platform. It is also important to note that the **Artist** inherits the role of a **Customer**, allowing them to use the same modules available to customers, such as purchasing artworks from other artists.

3.4 Functional and Non-Functional Requirements

3.4.1 Functional Requirements

Functional Requirements are the requirements that the user specifically demands as the basic facilities that the system should offer (GeeksforGeeks, 2025b). These requirements define what the system should do, how it should respond to particular inputs, and how it should behave under specific conditions. The tables below shows the functional requirements for each module in the Artist E-commerce system.

Table 3.1 Functional Requirements for Sales Module

Sales Module	
1.1	The system shall allow user to add artwork for sales
1.2	The system shall allow user to enter artwork name
1.3	The system shall allow user to enter artwork description
1.4	The system shall allow user to enter artwork price
1.5	The system shall allow user to upload artwork image
1.6	The system shall allow user to enter artwork stock quantity

1.7	The system shall allow user to change the artwork status to available or sold out
1.8	The system shall allow user to enter the category of the artwork
1.9	The system shall be able to validate user input before adding the artwork for sales
1.10	The system shall allow user to edit artwork details
1.11	The system shall validate the user information before update the details
1.12	The system shall allow user to delete artwork
1.13	The system shall confirm with the user before deleting the artwork
1.14	The system shall allow the user to view the reviews of the artwork
1.15	The system shall display review details such as rating, comment and image
1.15	The system shall display a list of uploaded artwork that the user sells
1.16	The system shall allow user to view their sales history and performance analytics

Table 3.2 Functional Requirements for Purchase Module

Purchase Module	
2.1	The system shall display artwork available for sales in a listing for user to select
2.2	The system shall allow user to view artwork details such as name, description, price, and stock quantity
2.3	The system shall allow user to search artwork based on name
2.4	The system shall allow user to filter artwork based on category
2.5	The system shall display recommended artwork based on user preferences
2.6	The system shall allow user to report the artwork if suspicious of fraud
2.7	The system shall allow user to rate and review the artwork after purchase
2.8	The system shall allow user to add the artwork to the shopping cart
2.9	The system shall allow user to set quantity to purchase the artwork
2.10	The system shall allow user to view the shopping cart
2.11	The system shall allow user to change quantity to purchase in the shopping cart
2.12	The system shall allow user to remove artwork from their shopping cart

2.13	The system shall validate quantity before updating
2.14	The system shall allow user to view the review and rating of the artwork
2.15	The system shall allow user to checkout their shopping cart
2.16	The system shall calculate the total amount to pay for the artwork in the shopping cart
2.17	The system shall allow user to enter shipping information
2.18	The system shall allow user to select payment option
2.19	The system shall validate the checkout detail before processing
2.20	The system shall confirm the user before checkout
2.21	The system shall allow user to apply discount on their purchase

Table 3.3 Functional Requirements for Artwork Fraud Detection Module

Artwork Fraud Detection Module	
3.1	The system shall scan artwork for sign of plagiarism, fraud, or AI generated images from database
3.2	The system shall flag artwork that has the sign of plagiarism, fraud, or AI generated images
3.3	The system shall notify user about the flagged artwork to review
3.4	The system shall allow users to view all the flagged artwork along with the reason for detection and supporting evidence
3.5	The system shall compare uploaded artworks with existing artwork records in the database to detect similarities
3.6	The system shall allow users to update the status of flagged artwork to Under Review, Verified or Rejected based on the outcome of the fraud detection process
3.7	The system shall notify the artist if there is any status changes on their artwork
3.8	The system shall generate fraud detection report summarizing the number of flagged artworks, resolved cases, and rejections for monitoring.
3.9	The system shall display statics about the fraud artwork for user to view

Table 3.4 Functional Requirements for Payment Module

Payment Module	
4.1	The system shall process the payment via third-party payment gateway
4.2	The system shall calculate the total amount of reward point should be given to the user when condition is met
4.3	The system shall generate digital receipt to user after payment process is successful
4.4	The system shall notify user payment successful or not successful
4.5	The system shall allow user to view their payment details
4.6	The system shall allow user to view the delivery status of their purchased artwork
4.7	The system shall allow users to review and rate the artwork after it arrived to the user shipping location

Table 3.5 Functional Requirements for AI Customer Service Chatbot Module

AI Customer Service Chatbot Module	
5.1	The system shall allow user to chat with chatbot for question such as delivery status, refund policy, or account issues.
5.2	The system shall process the user input to understand what they are asking
5.3	The system shall generate a response to the user based on their question
5.4	The system shall allow user to request for live support when user needed

3.4.2 Non-Functional Requirements

Non-Functional Requirements are the quality constraints that the system must satisfy (GeeksforGeeks, 2025b). Non-Functional Requirements describe how well the system performs its tasks and focuses on the performance, usability, and other quality attributes (GeeksforGeeks, 2025b). For the Artist E-commerce System, non-functional requirements ensure that the platform provides a better experience for both artists and customers. Meeting these requirements is important to improve the system architecture and overall performance of the system.

Reliability

Reliability specifies how likely the system would run without a failure for a given period of time under predefined conditions (AltexSoft, 2023). For Artist E-commerce, the system ***downtime should not be more than 0.05%***. This is important because even minimal disturbances can negatively impact user experience on the system. The system should also be able to handle and ***recover from unexpected errors or crashes without data loss or incorrect data processing***. This is because maintaining the integrity of transaction records, artwork listings, and user data which are critical to the reliability and smooth functioning of the system.

Usability

Usability indicates how effectively and easily a user can learn and use the system (AltexSoft, 2023). Artist E-commerce should have ***very low learning time, maintain consistency in every page, provide user guidance*** such as providing a tutorial setup, and ***use familiar icons*** such as using a trash icon for the delete button. This includes that the user should be able to ***easily access the most frequently used features with minimal navigation***. Additionally, users should also be able to ***find desired artwork products within three clicks from the homepage***.

Security

Security assures that all the data, such as customer details and transaction records inside the system, will be protected against malware attacks, data breaches, and unauthorized access (AltexSoft, 2023). Artist E-commerce ***should encrypt and securely store the user data***, including personal and payment information, to prevent any of the data from being leaked to any unauthorized personnel. Furthermore, all transactions should be ***processed through a secure and trusted payment gateway*** that is Payment Card Industry Data Security Standard (PCI DSS) compliant.

Performance

Performance defines how fast a software system or a component responds to certain users' actions under a certain workload (AltexSoft, 2023). In Artist E-commerce, the system shall load pages and ***respond to user actions within 3 seconds*** under normal conditions. Not only that, but the system should be able to ***support 1,000 concurrent users without degradation in service quality***. When searching or filtering the artwork listing, the system should also ***return results within 2 seconds***, even with large data sets.

3.5 Chapter Summary and Evaluation

In summary, this chapter summarizes the methodology and requirement analysis that form the requirements and functionality of the Artist E-Commerce System development. It is concluded that the methodology for the development of this project will be the Extreme Programming (XP) methodology due to its process and advantages, which align well with the needs of this project. Additionally, a detailed analysis of functional and non-functional requirements is provided to define the expected system behavior, performance, and constraints. These requirements were identified using requirement gathering techniques, which include document analysis and system reviews to ensure they align with real-world needs and academic insights. Overall, the methodology and requirement analysis establish a solid foundation for the system's design and successful implementation.

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